CUSTOMER SEGMENTATION

Business Intelligence & Machine Learning

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Executive Summary

This project looks at shopping patterns in a dataset of 3,900 customer records. I used Power BI to explore the data and build dashboards. I then used K-Prototypes to split customers into three groups based on a mix of number-based and category-based details. A Random Forest model learns to predict the group labels created by K-Prototypes. The app uses FastAPI to connect the model to the user interface, and Gemini writes a simple explanation of the result.

The file has 3,900 records and 18 columns, with one record for each CustomerID. The average purchase amount is about $255.48, and the total recorded purchase amount is about $996.38K. The three groups differ in spending and previous-purchase counts, but the Gower silhouette score (0.0676) suggests that the groups are not clearly separated. For that reason, they should be treated as an initial analysis, not as fixed customer types. The Random Forest accuracy of 95.26% shows how well it reproduces the labels created by K-Prototypes; it does not prove that the groups will improve business results.

1. Introduction

Customer segmentation means grouping customers who share some similar details or shopping habits. Businesses can use these groups to explore buying patterns and plan more relevant messages or offers. In this project, Power BI is used to explore the data, machine learning is used to create and predict group labels, and a small application presents the result.

2. Project Objectives

- Explore customer purchase patterns across product categories, purchase amounts, seasons, locations, discounts, and subscription status.
- Use mixed numerical and categorical customer features to create three exploratory customer groups with K-Prototypes.
- Train a Random Forest classifier to predict the cluster labels generated by K-Prototypes.
- Provide a FastAPI-based prediction workflow with a UI for customer inputs and a Gemini-generated plain-language explanation.
- Translate descriptive findings into cautious, testable business recommendations.

3. Dataset Description

The dataset contains 3,900 rows and 18 columns. It includes customer details, purchase amounts, product categories, subscription status, discount and promo_code use, and purchase frequency. Each CustomerID appears once, so this is a snapshot of customers not a record of every purchase they have made over time.

Attribute group	Columns / examples	Use in this project
Customer profile	CustomerID, Age, Gender, Location	Descriptive customer profile; CustomerID is an identifier, not a model feature.
Purchase details	ItemPurchased, Category, PurchaseAmountUSD, Size, Color, Season, ReviewRating	EDA and purchase-pattern analysis.
Engagement and promotion	SubscriptionStatus, DiscountApplied, PromoCodeUsed, PreviousPurchases, FrequencyOfPurchases	EDA and selected segmentation features.
Fulfilment/payment	ShippingType, PaymentMethod	Descriptive exploration; not included in the stated clustering feature list.

The data checks found no missing values or duplicate rows, and all 3,900 customer IDs were unique. However, the file does not include transaction dates, profit margins, customer lifetime value, or results from past marketing campaigns. This means the analysis cannot tell us whether a campaign caused a change, which group is most profitable, or how buying habits changed over time.

4. Methodology and Workflow

4.1 Power BI Exploratory Data Analysis

I used Power BI to summarise the records and make the patterns easier to see. The dashboard covers total and average purchase amounts, product categories, subscription status, season labels, locations, and discount use. These charts show what appears in this dataset; they do not prove that a discount, season, or location caused a particular result.

4.2 Feature Preparation

The stated K-Prototypes feature set includes the numerical attributes Age, PurchaseAmountUSD, and PreviousPurchases, and the categorical attributes Category, SubscriptionStatus, DiscountApplied, PromoCodeUsed, and FrequencyOfPurchases. CustomerID is excluded because it is an identifier. Numerical features were scaled before clustering; categorical values were retained as categorical features for the mixed-data clustering method.

4.3 K-Prototypes Clustering

I set K-Prototypes to create three groups using both numerical and category-based details. Each record receives a cluster number. The numbers simply identify the groups—they do not rank customers from best to worst.

4.4 Random Forest Classifier

I trained a Random Forest model to predict the group labels made by K-Prototypes. On the test set, it achieved 95.26% accuracy and a macro F1-score of 0.9531. This means it usually predicts the same labels that the clustering process created. It does not mean the model can predict future spending or guarantee better marketing results.

4.5 Application Layer: FastAPI and Gemini

In the app, a user enters details such as age, purchase amount, previous purchases, category, subscription status, and discount or promo-code use. FastAPI sends the inputs through the saved preprocessing steps and Random Forest model. The model returns a group label and class-score estimates. Gemini then explains the result in plain language. Gemini explains the prediction; it does not create the clusters or replace the Random Forest model.

5. Business Intelligence Findings

Measure / comparison	Reported result	Interpretation
Total purchase value	Approximately $996.38K	Aggregate purchase amount represented in the dataset.
Average purchase amount	Approximately $255.48	Average recorded purchase amount across 3,900 records.
Category totals	Clothing $209K; Accessories $208K; Home Decor $197K; Electronics $194K; Sports $189K	Category purchase totals are relatively close; Clothing is highest in the reported dashboard.
Average purchase by category	Clothing $262.28; Home Decor $259.21; Accessories $253.74; Sports $251.14; Electronics $250.88	Average purchase amount varies modestly by category.
Subscription status	49.51% Yes; 50.49% No	The dataset is nearly evenly split between subscribed and non-subscribed customers.
Season labels	Spring $272K; Fall $253K; Winter	Descriptive comparison by supplied season label only; no transaction dates are

	$242K; Summer $226K	available to establish temporal trends.
Location record counts	New York 812; Los Angeles 806; Chicago 777; Houston 764; Miami 741	Record counts are similar across the five listed locations.

5.1 Discount Analysis

The dashboard lets us compare records with and without discounts or promo codes. But this dataset does not show profit margins or results from a controlled test, so we cannot say that discounts caused higher spending or increased profit. Before changing a discount policy, a business should test the offer and check both sales and margin.

6. K-Prototypes Segment Profiles

Cluster	Customers	Average purchase amount	Median purchase amount	Average previous purchases
Cluster 0	1,488	$111.27	$105.88	6.67
Cluster 1	1,220	$337.86	$344.54	11.03
Cluster 2	1,192	$351.20	$353.47	3.07

Feature	Cluster 0	Cluster 1	Cluster 2
Most common category	Accessories (23.2%)	Clothing (24.9%)	Electronics (21.7%)
Subscription mode	No (56.3%)	Yes (53.6%)	Yes (52.6%)
Discount applied mode	No (53.9%)	Yes (54.3%)	No (54.0%)
Promo code used mode	No (55.6%)	Yes (57.4%)	No (55.3%)
Purchase frequency mode	Monthly (39.0%)	Weekly (39.9%)	Fortnightly (38.3%)

The biggest differences between the groups are average purchase amount and previous-purchase count. Cluster 0 has a much lower average purchase amount than Clusters 1 and 2. Cluster 1 has the highest average previous-purchase count, while Cluster 2 has the lowest. The most common categories and engagement patterns are less distinct, and the overall separation score is low. These are early observations, not firm customer profiles.

7. Cluster Quality and Limitations

The Gower silhouette score was 0.0676, calculated on a sample of 600 records. This score is close to zero, which suggests that many customers do not fit clearly into one group rather than another. The groups can still be explored, but they need more testing before a business relies on them.

- The selected features may not distinguish customers strongly enough, and several category/engagement modes are similar across groups.
- Purchase amount separates Cluster 0 from the other groups more clearly than many categorical features, so spending may drive much of the difference.
- The dataset is a single snapshot and lacks transaction dates, cost/margin, customer lifetime value, and measured campaign outcomes.
- The number of clusters (K = 3) is a project choice; additional K values and feature combinations should be compared before operational use.
- A high Random Forest score against generated cluster labels shows label reproduction, not that the segmentation is useful for business decisions.

8. Application Architecture

The project has two main parts. Power BI is used to explore the dataset and show the findings. The machine-learning part prepares the selected features, creates groups, trains a model to predict those group labels, and makes the prediction available through the app.

Component	Role
Customer dataset	Source of the 3,900 customer records.
Power BI	EDA and descriptive dashboards; not the clustering engine.
Processing	Selects model features and prepares numeric/categorical inputs.
K-Prototypes	Creates three cluster labels from mixed data.
Random Forest	Learns to predict the generated cluster labels.
Saved model artifacts	Persisted preprocessing/model components used during inference, according to the implemented pipeline.
FastAPI	Backend endpoint that receives inputs and returns a prediction.
User interface	Collects customer attributes and displays the segment result.
Gemini API	Produces a plain-language explanation based on the prediction context and submitted inputs.

When someone submits the form, the app sends the details to FastAPI. The saved preprocessing steps and Random Forest model prepare the input and predict a group. Gemini then writes a short explanation, and the app displays both the group and the explanation. During training, K-Prototypes creates the labels that the Random Forest learns to predict.

9. Application UI

9.1 Customer Input Screen

The first screen lets a user enter the customer details needed by the model. After submitting the form, the user can see the predicted group without running Python code manually.

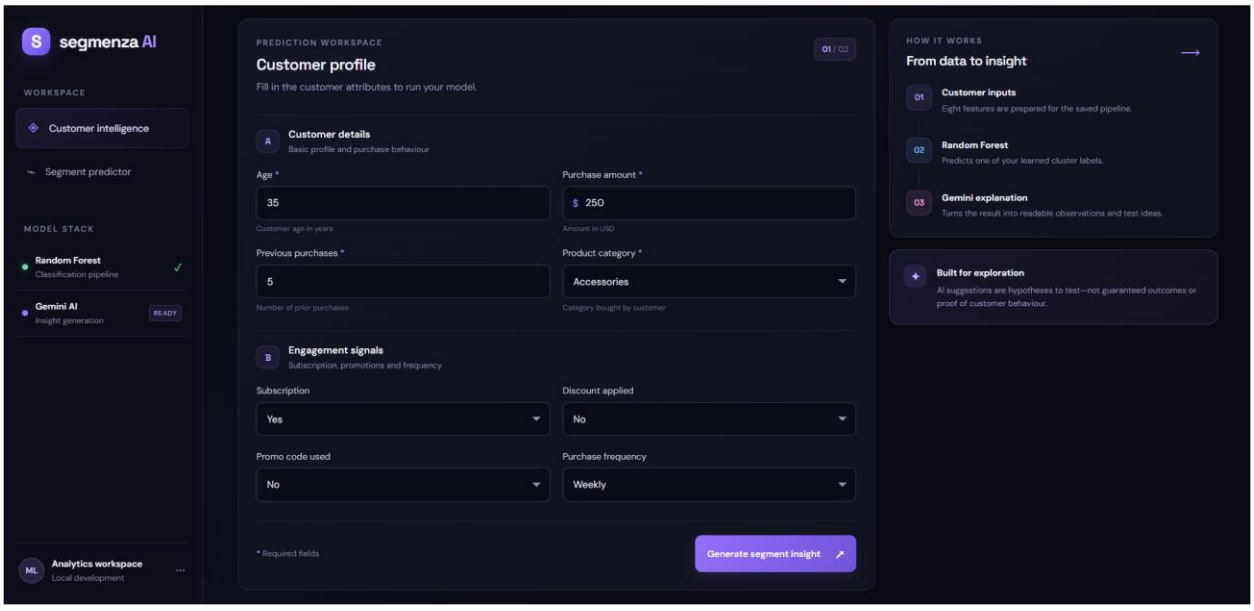


Figure 1. Customer profile and engagement inputs in the application UI.

9.2 Prediction and Explanation Screen

The results screen shows the predicted group, the model's class-score estimates, and an explanation written by Gemini. The example shown in the screenshot predicts Cluster 2 for the entered details. This is only one example. Cluster numbers are labels, not rankings.

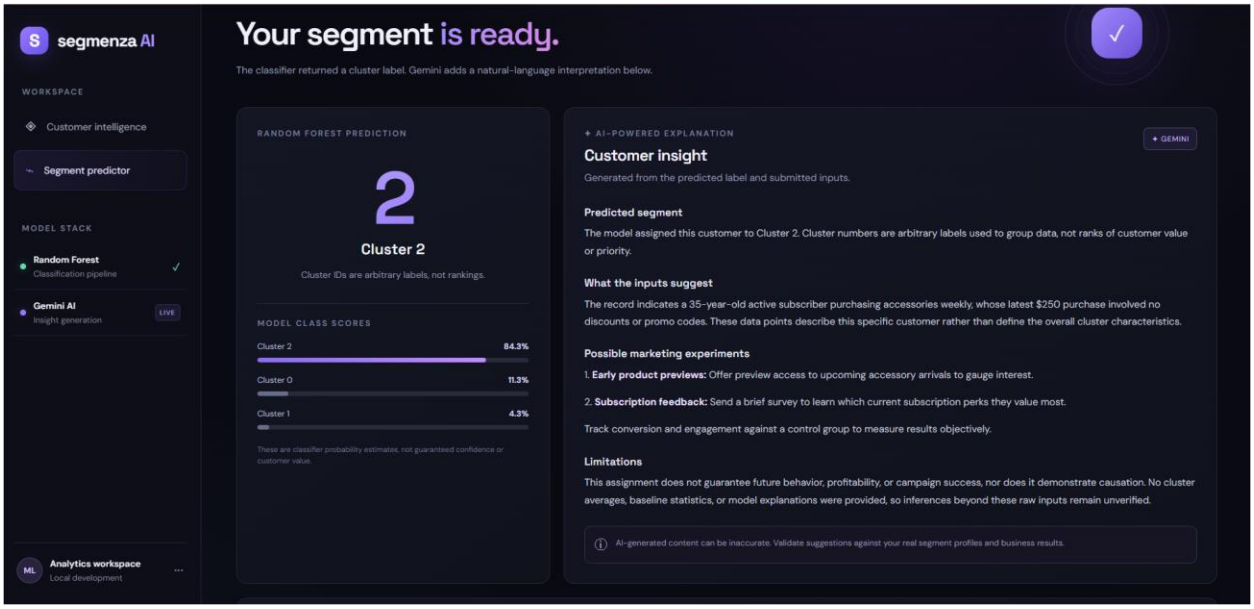


Figure 2. Example prediction result and AI-generated explanation.

10. Business Suggestions

The ideas below are practical options to test, not guaranteed results. The dataset shows purchase patterns, but it does not include profit margins or past campaign results.

- Start with small tests instead of changing the whole marketing plan at once. For example, try one offer with a small customer group and compare the results with a similar group that did not receive it.
- For Cluster 0, explore simple ways to encourage a larger basket, such as product bundles or recommendations for related items. Check whether the average order value improves before expanding the offer.
- For Cluster 1, customers have the highest average number of previous purchases and weekly frequency is common. Test a loyalty benefit or a subscription reminder, then measure repeat purchases and opt-outs.
- For Cluster 2, the average purchase amount is high but the average number of previous purchases is low. Try relevant product recommendations or a follow-up message, but do not assume these customers are new or about to leave—the dataset cannot confirm that.
- Do not offer discounts to everyone by default. Test different offers and track conversion, total sales, and gross margin together. A higher purchase amount alone does not mean the offer made more money.
- Use the category findings to plan what to test, not to make a final stock or pricing decision. Clothing and Accessories have the highest recorded category totals in this dataset, but the differences are small and the file does not include profit or stock data.
- Before using the three groups in a real campaign, compare other cluster settings and ask whether the groups lead to actions that are genuinely different. The current silhouette score is low, so it is worth checking if a simpler rule-based grouping works just as well.
- Review the results after each test and keep only the changes that improve the chosen business measure. Record the test dates, audience, offer, sales, and margin so future analysis can be more reliable.

11. Conclusion

This project brings together a Power BI dashboard, K-Prototypes clustering, a Random Forest classifier, and a FastAPI app with Gemini explanations. The dashboard summarises purchase amounts, categories, locations, season labels, and subscription status. The three groups differ in spending and previous-purchase counts, but the low silhouette score shows that they are not clearly separated. The project is a useful starting point for exploring customer segmentation, but the groups should be tested further before they are used to make business decisions.

Appendix A. Key Results at a Glance

Item	Result
Dataset size	3,900 records; 18 columns; one row per unique customer ID
Total purchase value	$996.38K
Average purchase amount	$255.48
K-Prototypes	K = 3
Cluster sizes	C0: 1,488; C1: 1,220; C2: 1,192
Gower silhouette	0.0676 on sample of 600
Random Forest test accuracy	95.26% against K-Prototypes-generated labels
Random Forest macro F1	0.9531 against K-Prototypes-generated labels
Application	FastAPI + UI + Gemini explanation layer

Chapter 12 Power BI Dashboard and Graphical Analysis

This chapter includes the Power BI dashboard and the charts used to explore the data. The visuals summarise the records we have; they do not prove why customers bought something or whether a business action worked.

Power BI dashboard overview

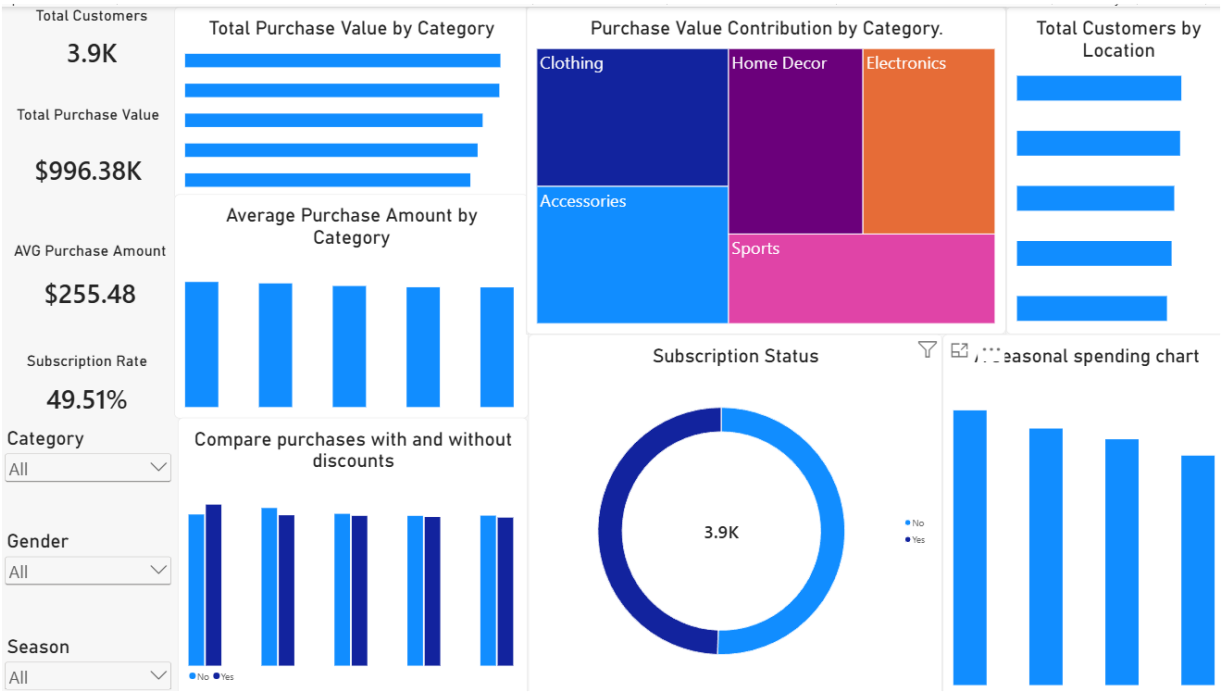


Figure 1. Power BI dashboard overview.

Customer purchase overview

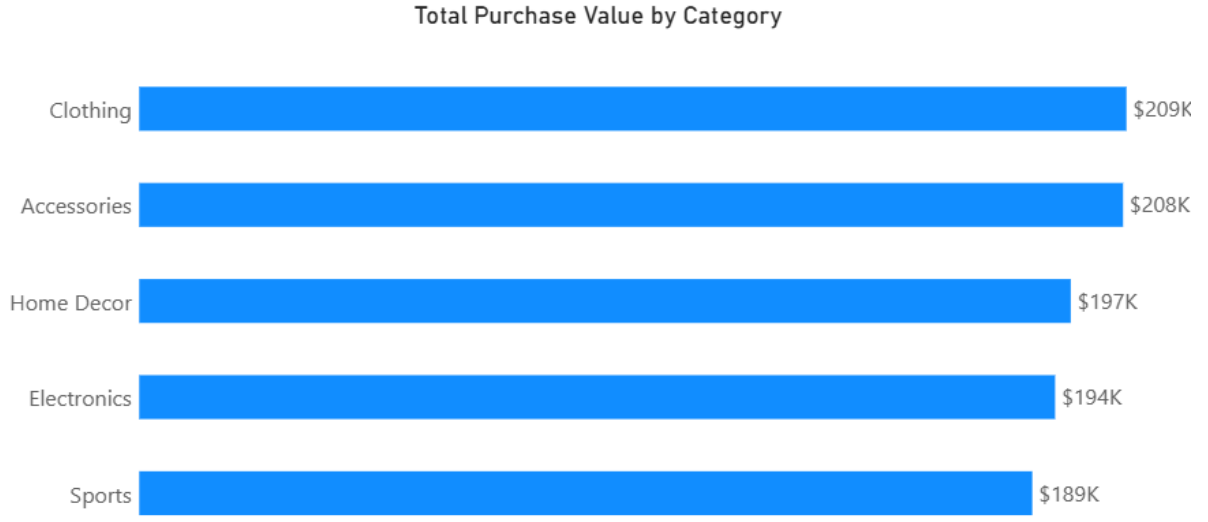


Figure 2. Customer purchase overview.

Purchase amount and category analysis

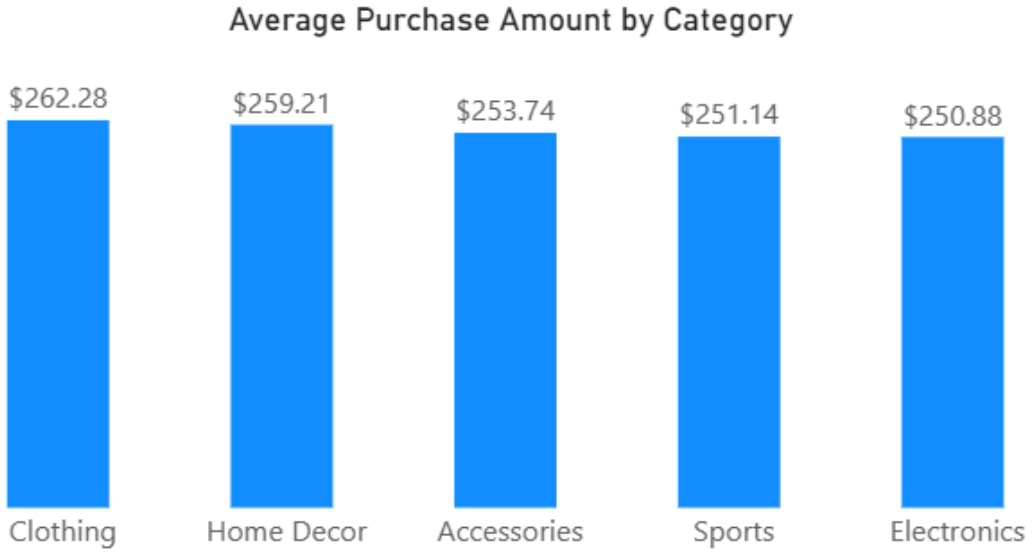


Figure 3. Purchase amount and category analysis.

Customer distribution and purchase patterns

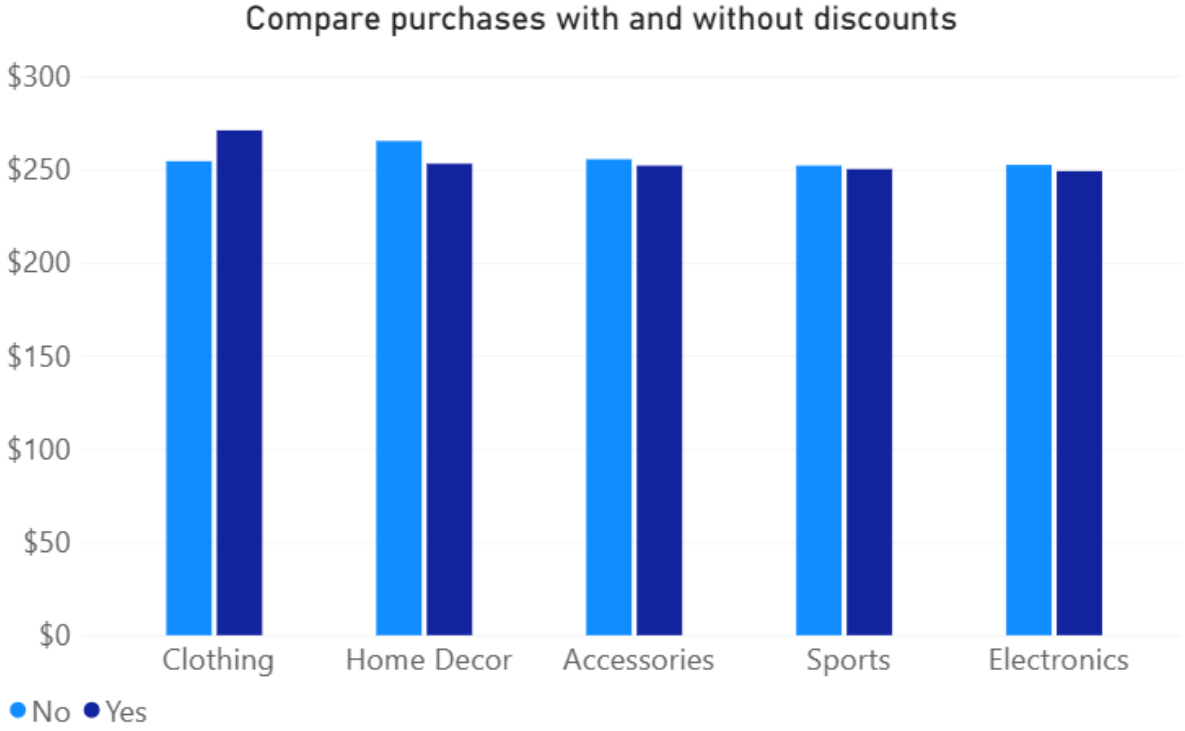


Figure 4. Customer distribution and purchase patterns.

Discount and promotion comparisons

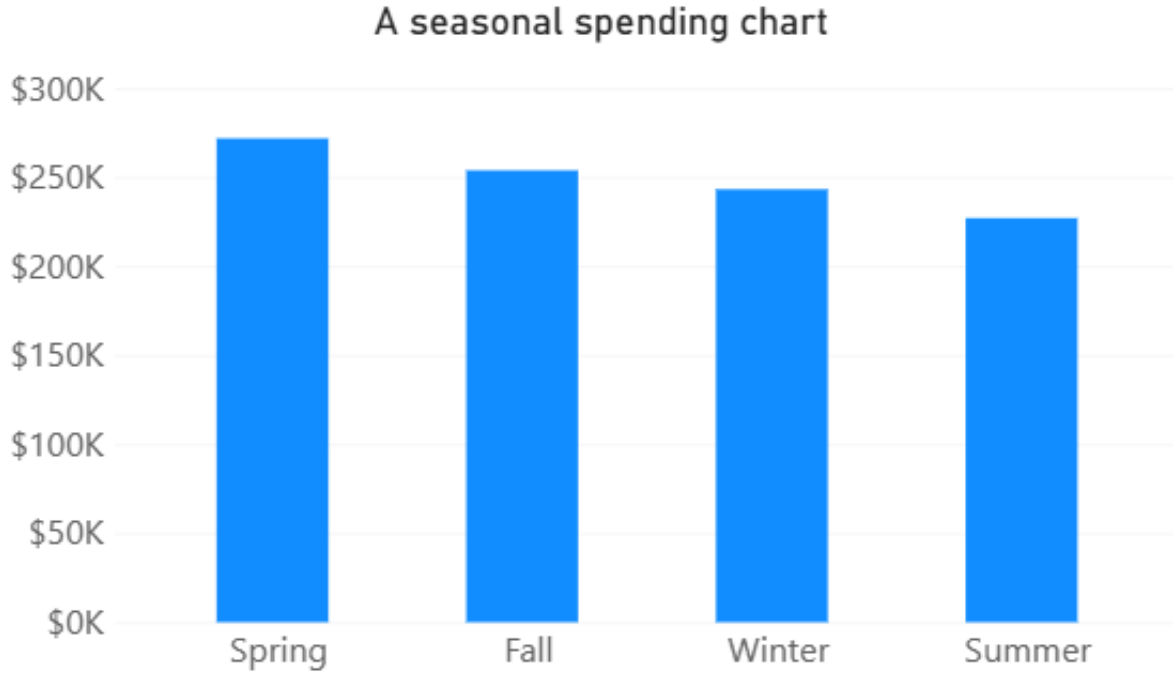


Figure 5. Discount and promotion comparisons.

Seasonal and location analysis

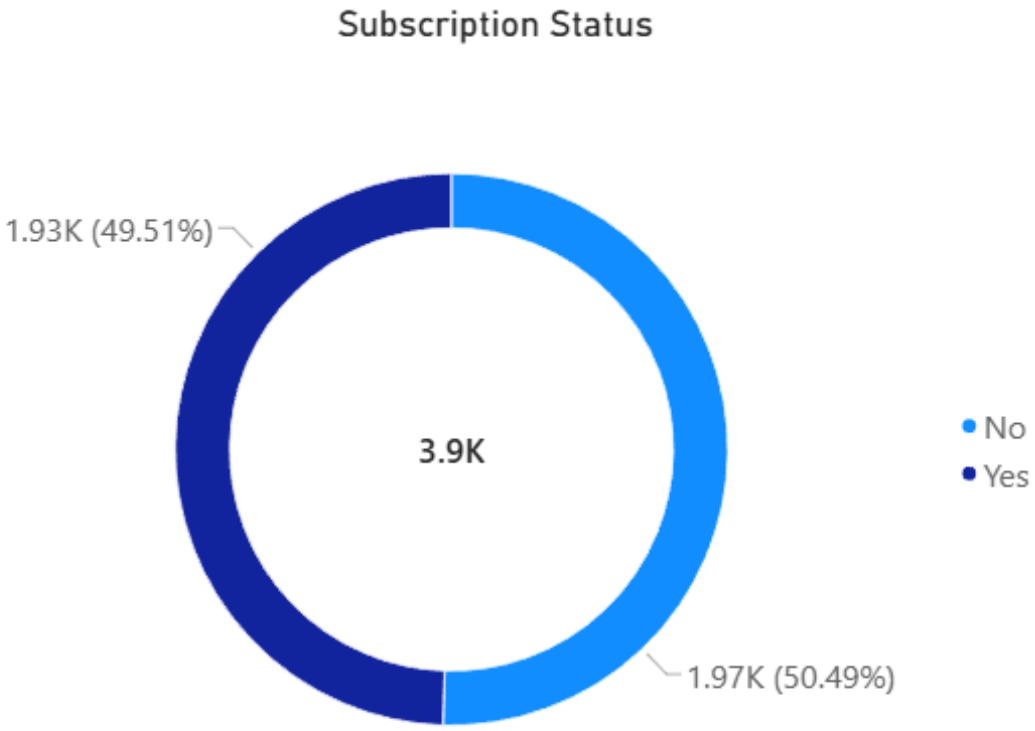


Figure 6. Seasonal and location analysis.

Subscription and category analysis

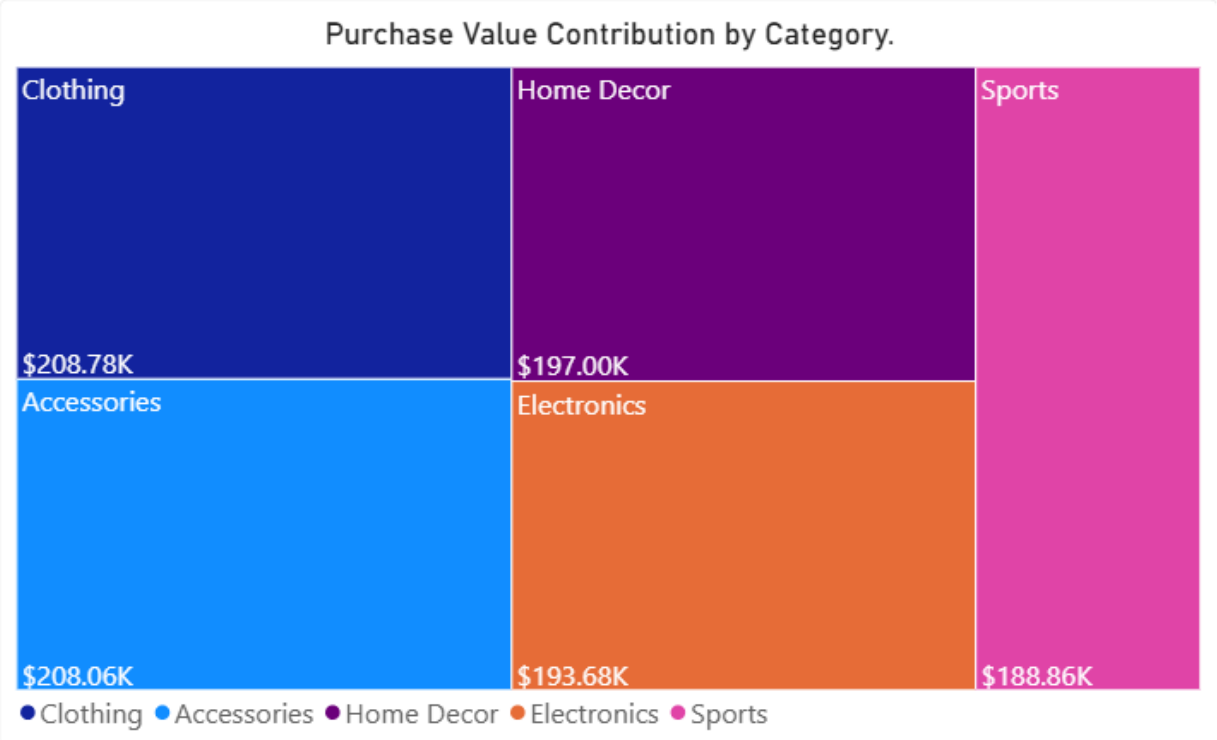


Figure 7. Subscription and category analysis.

Additional Power BI analysis view



Figure 8. Additional Power BI analysis view.

Key observations from the dashboard

- Total purchase value is approximately $996.38K, and average purchase amount is approximately $255.48.
- Clothing has the highest reported category total at about $209K, followed closely by Accessories at about $208K.
- Average purchase amount ranges from about $250.88 for Electronics to $262.28 for Clothing.
- Subscription status is nearly evenly split: 49.51% Yes and 50.49% No.
- Season-label totals are Spring $272K, Fall $253K, Winter $242K, and Summer $226K. These are label comparisons, not time-series trends.
- Location counts are New York 812, Los Angeles 806, Chicago 777, Houston 764, and Miami 741.
- Discount and promo-code comparisons are descriptive only; they cannot establish causal impact or profitability without controlled testing and margin data.